

# FEATURES





# Don't ask, just listen...

## Using social networks to measure subjective well-being

How do people feel about their lives and the societies in which they live? Are they happy, hopeful or concerned about the future? Surveys can help answer these questions, but **Stefano M. Iacus** and **Giuseppe Porro** argue for using social networks and sentiment analysis to let citizens speak for themselves

Governments have long sought to assess the state of society using indicators such as gross domestic product (GDP) and employment rates. During the Covid-19 pandemic, for example, these metrics told a clear story about the economic downturn experienced by many countries. But what about the well-being of citizens at the time? How did people feel about orders to “stay at home”? How did they cope with the deaths of friends and relatives, or the loss of jobs? Were they confident, concerned, or scared about their futures?

Getting timely and reliable answers to questions like these can be difficult at the best of times. Surveys are the usual method by which people will be asked about their

thoughts and feelings on various issues, but survey response rates have declined over time, meaning that it has become harder to find people to answer questions. In addition, the order of questions in a survey can influence the answers given, a problem seen by economists Angus Deaton and Arthur Stone in their study of a survey on well-being.<sup>1</sup> There can also be a lag between an event happening and a survey respondent being asked about it, and this can bias the answers that are given – it can, after all, be difficult to remember how we felt about something that happened in the past.

For these reasons, and more, we believe there is a need to monitor opinion and sentiment within communities in real time, ►

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- in order to help inform public policies and assess their outcomes. To address this need, we can look to social media as a source of data on what members of the public are thinking and feeling.

### The “virtual agora” as data source

The great advantage of social networking sites as data sources is that users share their reactions to specific events without a researcher needing to explicitly ask a question. The researcher need only *listen* to what users are saying and infer from their words information about mood and emotion. The expressions of feelings and opinions are spontaneous and not encouraged in any way, hence they are more likely to reflect the actual mindset of their authors, or at least how they want to be perceived.

Provided users of platforms such as Facebook and Twitter set their accounts to be public, and social media posts are analysed on an aggregated and anonymised basis, it is possible to monitor opinion and sentiment continuously, at relatively low cost, and without violating privacy rights.

But there are caveats. The main objection to using social media data would be that users of these sites are not a representative subset of the population: they may be younger and more tech-savvy, have a different education level, and be heavily concentrated in particular parts of a country, such as large urban areas. However, official surveys that collect data about the socio-demographic features of internet and social media users make it possible to weight statistics based on social media data in order to get closer to a “representative” estimate of the feelings of the entire population. And, perhaps more importantly, the opinions of internet and social media users as a distinct group are significant *per se*: their posts and conversations can be read by a wide set of people – including journalists, who often quote social media comments in their stories – and so this group may exert a sort of “lighthouse effect”, in that they

may anticipate, affect and shape public opinion more broadly. For this reason, the importance of social media as an information source cannot be ignored.

### How to extract sentiment from tweets

Social media data is big data, and the huge amount of information contained within tweets and Facebook posts cannot be decoded and separated from background noise without the help of appropriate statistical techniques.

Many methods have been developed to overcome the problem of extracting meaning and sentiment from an enormous amount of text. Most automate the task, and these are known as *unsupervised methods* of sentiment analysis, with *sentiment analysis* being the process of determining the mood expressed by a collection of texts. Typically, a dictionary is created in which words are associated with different feelings (joy, fear, sadness, etc.). A naive algorithm then looks for the presence of these words in the texts being analysed,

and then flags parts of the texts as conveying particular feelings, depending on the words contained within.

Unsupervised methods are powerful in that they can classify large data sets with very low effort. But they often struggle when analysing texts that make use of paradoxes, irony, neologisms or swear words, as can be the case with social media conversations.

An alternative to unsupervised methods are *human-supervised methods*. These perform the same task of sentiment analysis but without relying on predefined dictionaries of terms. In this case, a sample of texts are read and classified by human coders. Information from the human coders is then used to train an algorithm, and the algorithm learns how to combine language and mood in order to classify a larger set of texts. In other words, the relationship between words and sentiment is not assumed *ex ante*, but is inferred from the texts themselves.

As described in our recent book,<sup>2</sup> and with the help of the (human-supervised) *integrated Sentiment Analysis* (iSA) algorithm,<sup>3</sup> we set out



**Figure 1:** An example of a tweet and how it has been manually classified to train the iSA algorithm: positive for the components “emotions” and “resilience”, and negative for the component “vitality”.<sup>2</sup>



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to evaluate people's well-being in two countries – Italy and Japan – using posts collected from Twitter. Our research group chose to compare the Twitter populations of these two countries because it was scientifically challenging in terms of textual analysis, given the different types of writing used. Thus, it presented a good exercise to test the iSA algorithm, which was designed to be language agnostic. We started to analyse Twitter data for Italy in 2012, and for Japan in 2015.

To illustrate our approach to sentiment analysis, we will show how subjective well-being, captured through Twitter data, has evolved during the Covid-19 pandemic within these two distinct communities. We will also discuss a method to identify factors related to the pandemic that may potentially have an impact on well-being.

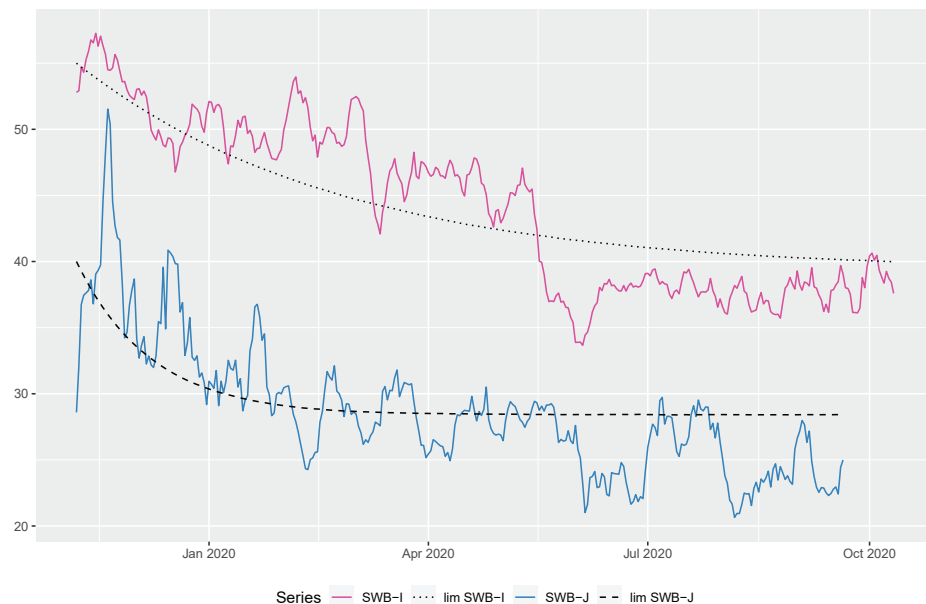
## Subjective well-being

Evaluating individual and collective well-being has become a popular and ambitious goal of social scientists and policy-makers in recent years, in recognition of the fact that GDP – the typical measure of a country's growth and prosperity – tells us little about the overall state of society.<sup>4</sup>

Alongside *objective* indicators that are able to take account of dimensions of social and economic prosperity that are neglected by GDP (such as sustainability), researchers are also focusing on measures of *subjective* (i.e., self-perceived) well-being, in order to capture components of well-being that are not possible to directly observe through objective variables.

It should be emphasised that the notion of subjective well-being is multifaceted: *emotional* (or hedonic) well-being must be distinguished from *evaluative* well-being (or life satisfaction), and from *eudaimonic* well-being (related to the sense of purpose and meaning in life) as summarised in the OECD guidelines.<sup>5</sup> Social media data seem especially suited for estimating and monitoring emotional well-being, in that social media users express their own transient feelings more often than their own overall life evaluation.

To test out this idea, in our research group we calculated a subjective well-being (SWB) index for both Italy (SWB-I) and Japan (SWB-J). The SWB-I and SWB-J indices consist of eight dimensions representing



**Figure 2:** The daily oscillation of SWB-I and SWB-J from November 2019 to 10 October 2020 for Italy and 20 September for Japan, and their trend estimates ("lim SWB-I" and "lim SWB-J"). The vertical axis represents the value of the index on the [0-100]% scale. The total volume of tweets is 13,975,242 for Italy and 12,907,902 for Japan.

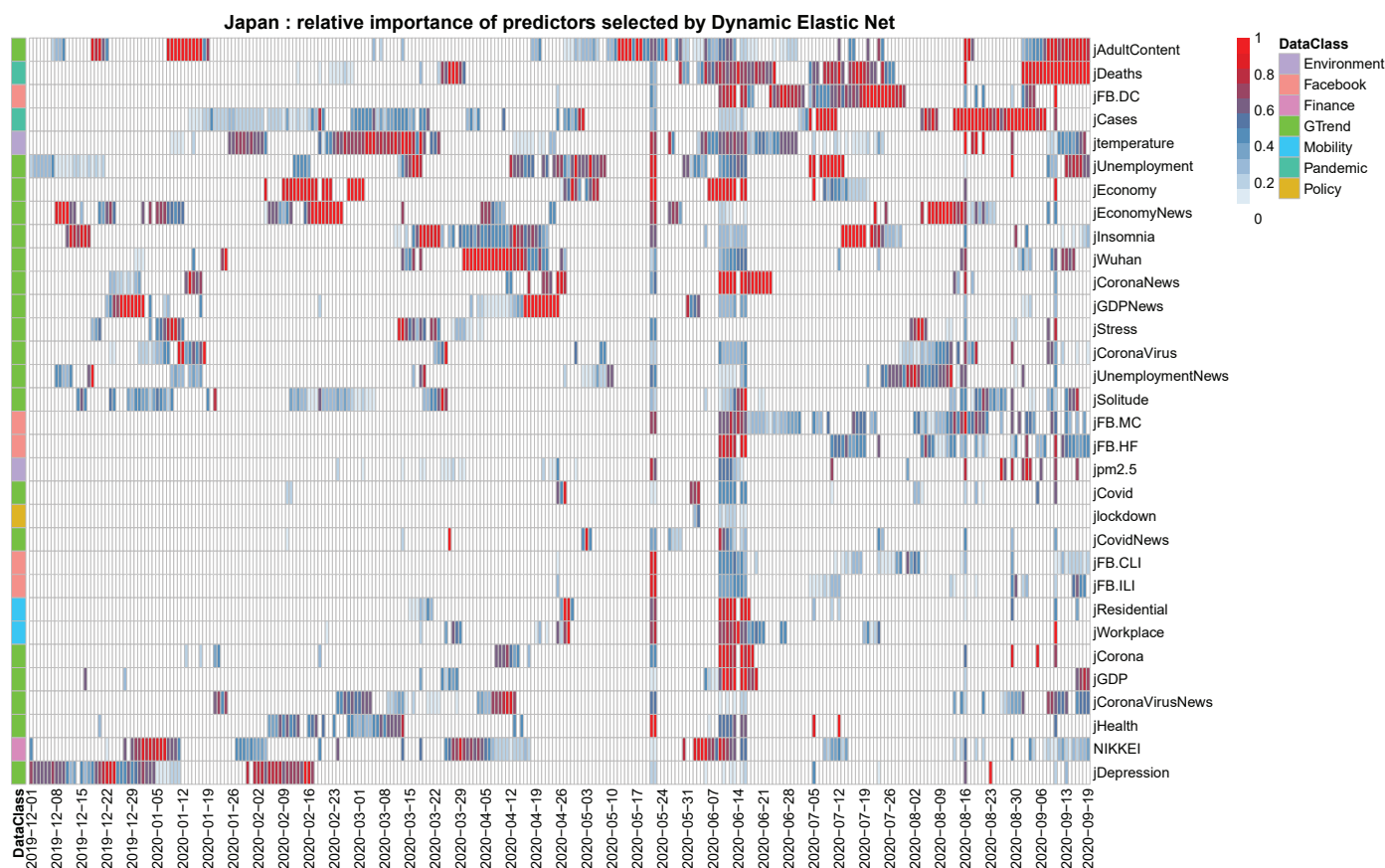
three different areas of well-being: personal (*emotions, life satisfaction, vitality, resilience, functioning*), social (*trust and belonging, quality of relationships*) and well-being at work (*satisfaction at work*). The basic process is as follows. First, the semantic content of Twitter messages is extracted by the iSA algorithm after a sample of texts have been manually classified as either "negative", "neutral", "positive", or "off-topic" according to each of the eight dimensions. For example, a tweet like the one in Figure 1 has been classified as positive for "emotions" and "resilience", and negative for "vitality". Then the iSA algorithm estimates the distribution of positive, negative, neutral and off-topic categories for all tweets, for each of the eight dimensions. Indices are then calculated – and rescaled to the interval [0-100]% – as the proportion of positive sentiment over the total of positive and negative sentiment, disregarding the neutral and off topic categories. The SWB-I and SWB-J indices are then the averages of these eight indicators.

If we examine SWB-I and SWB-J from November 2019 to September/October 2020 (see Figure 2), what leaps out first is that self-perceived (or self-declared) well-being is systematically higher in Italy than

## Our well-being indices have in common a significant drop around the point where Covid-19 emerges

in Japan. This difference may be attributed to specificities in the way the two cultures express emotions and feelings. (It has been observed that Asian countries tend to record lower scores in reported subjective well-being than continental European or Anglo-Saxon countries.<sup>6</sup>) Therefore, the absolute value of SWB-I in comparison to SWB-J does not tell us much. What is more important is the relative variation in the indices and their trends over time.

Both have in common a significant drop around the point where Covid-19 emerges, in late 2019 and early 2020. This decrease in the well-being indices was expected, given the spread of infection, restrictions, increasing numbers of deaths, and the impact of the virus on perceptions of daily life and expectations for the future. Figure 2 also suggests that the decrease in well-being due to the pandemic was not merely a temporary blip: as of October 2020, subjective well-



**Figure 3:** The relative importance of each variable through time in predicting the value of the SWB-J index (1 = max importance, 0 or no colour = not important) through a dynamic elastic net model.

- being appeared to have settled to a new lower level for both countries.

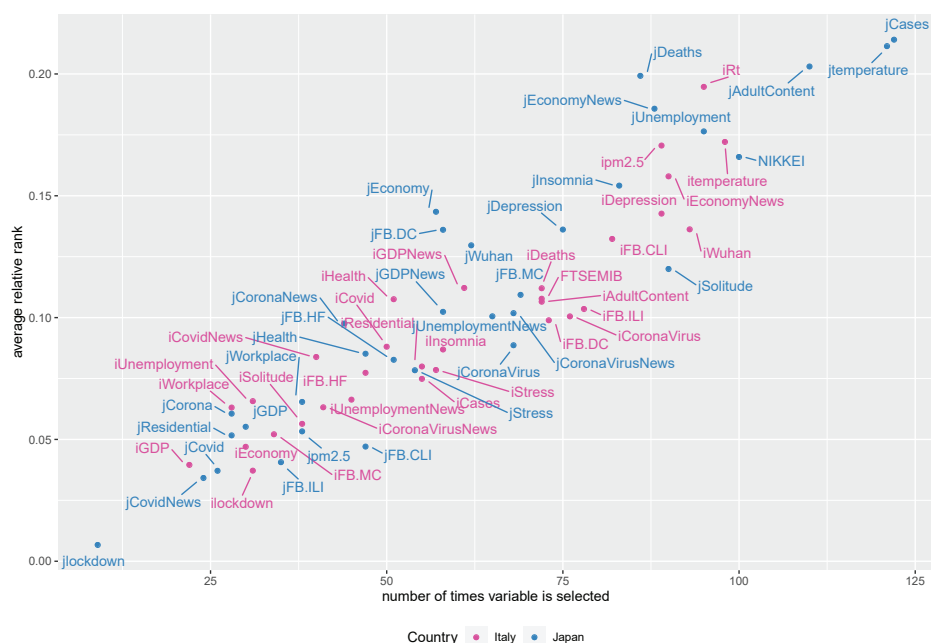
We can extend our analysis of subjective well-being to include an investigation of the factors that may have impacted the value of SWB-I and SWB-J during the pandemic. Making use of a dynamic elastic net model – an estimation method that performs estimation and model selection at the same time, applied to a moving time window – we can extract the temporal pattern of factors that statistically correlate with the index variations. Figure 3 is a graphical representation of the resulting analysis for the SWB-J index. It shows at what point in time a factor impacted on subjective well-being as well as the relative importance of this impact. As external factors we considered epidemiological data (deaths, cases), Google search topics (coronavirus/ Covid, stress, depression, solitude, etc.),

financial market indices (Nikkei, FTSE-MIB), GDP, unemployment rate, adult content consumption, temperature, and pollution levels (PM2.5), a few Facebook survey items about Covid or flu-like symptoms, and policy restriction indicators.

Figure 3 shows, for example, that the number of Covid cases and deaths (located at the top right of the y-axis, labelled jCases and jDeaths), as well as online adult content consumption (jAdultContent), had a substantial impact on well-being from July 2020, while news about Wuhan and GDP (jWuhan and jGDPNews) were quite important

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earlier on in the pandemic timeline, around March and April. But this is quite a complex display, and it is not simple to summarise and interpret. Figure 4 shows a summary version of the same data, comparing Italy and Japan. It shows the number of days each factor was selected by the model, plotted against the average relative importance of each factor in explaining the outcome values of SWB-I and SWB-J. (Importance is calculated via a random forest algorithm.) Factors that appear in the top right-hand corner of the figure are the most important ones in influencing – and also predicting – the well-being index value; those in the bottom left-hand corner are factors whose impact on well-being is less significant. For example, Covid’s reported reproduction number had a relatively high impact on perceived well-being in Italy (iRt). One possible explanation for this is that the national media amplified



**Figure 4:** Summary of relative importance (average relative rank) and number of times each variable is selected (over 315 dates for Italy and 294 dates for Japan) by the dynamic elastic net model.

the discussion about this particular epidemiological metric. Meanwhile, in Japan, Covid-19 cases and deaths were among the top factors impacting on well-being. We can also see that lockdown measures had a greater (though not substantial) impact on well-being in Italy (i lockdown) as compared with Japan (j lockdown), probably because of the different approaches adopted by the two countries (for example, Japanese law does not allow for enforced lockdowns (nyti.ms/3rIJSwL)). And we can also see that news about the state of the economy and unemployment were more relevant for Japan (jEconomyNews and jUnemployment) than for Italy (iEconomyNews and iUnemployment).

## The best of both

Rapid advancement of both computational power and analytical methods has made available an increasing number of well-being indicators based on social media data. The mutual compatibility of these indicators is still to be clarified – different indicators can give very different evaluations of well-being, even in the same communities. Also, additional effort must be made to study possible relationships among the emotional

statuses expressed by social media texts and the evaluative (life satisfaction) and eudaimonic (sense of purpose) dimensions of well-being. However, it cannot be denied that social scientists and policy-makers now have a much wider range of channels at their disposal for understanding individual and collective well-being, and these instruments can be used not only to keep policy-makers informed, but also to influence well-being through active interventions.<sup>7</sup>

A primary task to be performed, and with urgency, is to reconcile these new sources of information with official statistics. Established economic indicators such as GDP and unemployment data may have limitations and inconsistencies as measures of a country's growth and progress, but we should not neglect them in our attempts to understand well-being: a community experiencing a prolonged period of high unemployment, for example, may experience a decline in measured subjective well-being over time.

There are benefits to be had in bringing these two data sources together. Official data on unemployment come, at best, quarterly, and such data usually have a high level of territorial aggregation and

they ignore the impact that working environments and professional relationships can have on personal and social well-being. Therefore, anchoring a social media-based indicator of well-being at work to the official unemployment time series may help to augment the information the official statistics convey, in terms of space-time granularity and the qualitative features of the phenomenon. The official measures may also help to correct some of the bias in representation that is a feature of internet and social media data, giving these new sets of tools greater reliability.

Social media data is not a straight substitute for official statistics and traditional metrics. Rather, the opportunity before us is to use social media data to enrich the information available to policy-makers and social scientists, giving them more high-frequency, multifaceted, disaggregated, flexible analytical tools. ■

## Note

Stefano M. Iacus and Giuseppe Porro are the authors of *Subjective Well-Being and Social Media*, published in 2021 by CRC Press.<sup>2</sup>

## Disclosure statement

The authors declare no conflicts of interest.

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